Phase III: Software Design and Modeling

1. Software Architecture

System Architecture

1. **Citizen registers an account**

    - Request goes to Backend
    - Backend validates input (email uniqueness, password format)
    - Backend encrypts password
    - Data stored in the Users table
    - Citizen account created successfully

2. **Citizen logs into the system**

    - Request goes to Backend
    - Backend verifies email and password
    - Backend checks encrypted password in Users table
    - If valid, session is created
    - Citizen is redirected to dashboard

3. **Citizen browses sports facilities**

    - Request goes to Backend
    - Backend queries the Facilities table
    - Backend returns list of facilities
    - Citizen sees available facilities with details

4. **Citizen checks facility availability**

    - Request goes to Backend
    - Backend queries:
        - facility_schedules table (working hours)
        - bookings table (existing reservations)
    - Backend calculates available time slots
    - Backend returns available slots
    - Citizen sees real-time availability

5. **Citizen books a facility**

    - Request goes to Backend
    - Backend validates citizen session and selected time slot
    - Backend checks for conflicts in Bookings table

- If slot is free:
    - Booking is created
    - Stored in Bookings table
- Confirmation sent to citizen

6. **Citizen modifies a booking**

    - Request goes to Backend
    - Backend verifies booking ownership
    - Backend checks new time slot availability
    - If valid:
        - Booking is updated in Bookings table
    - Updated confirmation returned

7. **Citizen cancels a booking**

    - Request goes to Backend
    - Backend verifies booking
    - Booking status updated to cancelled in Bookings table
    - Slot becomes available again

8. **System prevents double booking**

    - When booking request is received
    - Backend checks existing records in Bookings table
    - Unique constraint ensures no duplicate time slot per facility
    - If conflict found
    - Booking rejected
    - Error returned to citizen

9. **Citizen receives booking notifications**

    - After booking action
    - Backend triggers notification service
    - Confirmation/reminder sent to citizen (email or system notification)

10. **Administrator adds or manages facilities**

    - Request goes to Backend
    - Backend verifies admin role

- Backend inserts/updates/deletes data in Facilities table
- Changes saved
- Updated facilities visible to citizens

11. **Administrator manages citizens**

- Request goes to Backend
- Backend checks admin privileges
- Backend updates citizen roles or data in Users table
- Changes applied successfully

12. **Administrator views reports and statistics**

- Request goes to Backend
- Backend queries Bookings, Users, and Facilities tables
- Aggregated data returned
- Reports displayed

13. **Facility staff verifies bookings**

- Request goes to Backend
- Backend validates staff role
- Backend updates booking status (verified) in Bookings table
- Booking marked as confirmed

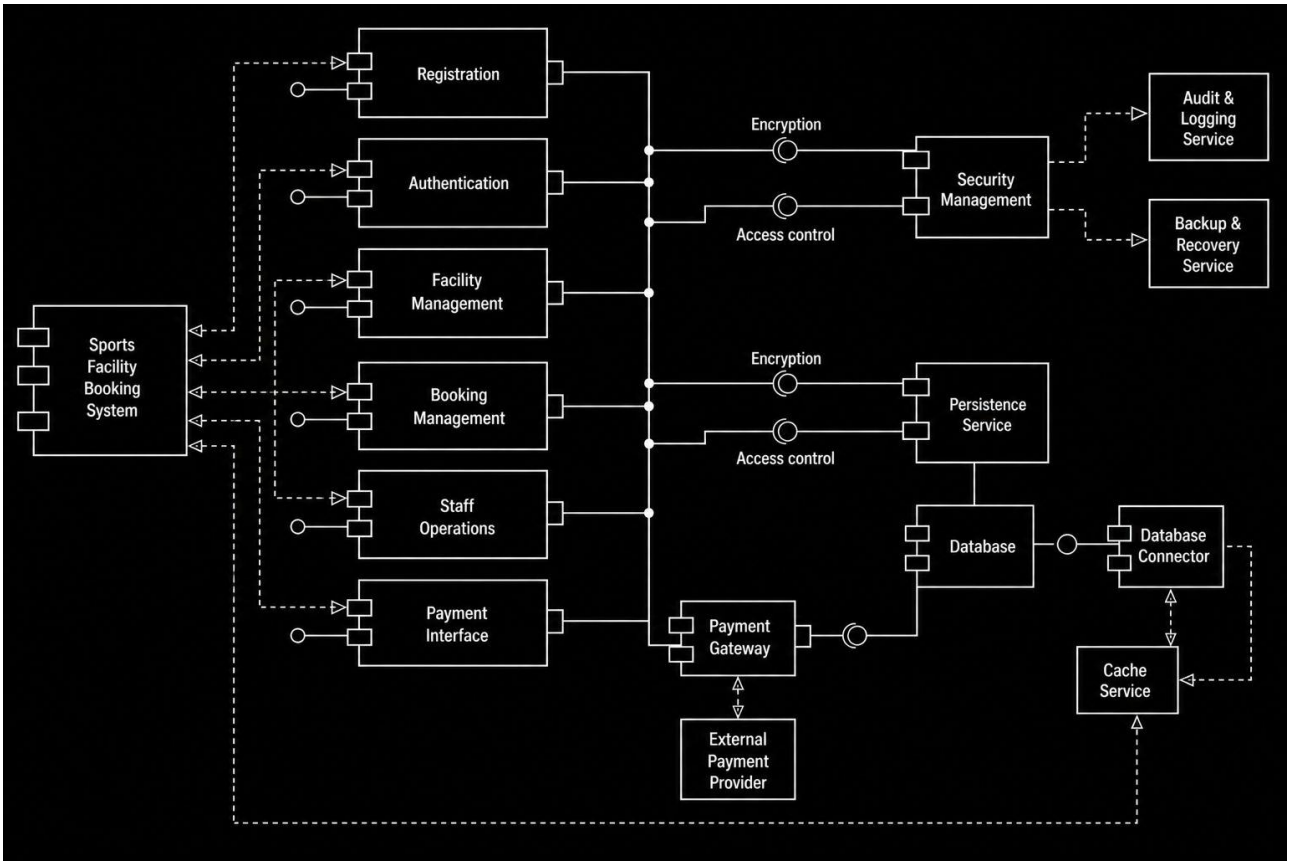
14. **Facility staff views daily schedule**

- Request goes to Backend
- Backend queries Bookings table filtered by date and facility
- Backend returns schedule
- Staff sees daily bookings

15. **Citizen makes payment**

- Request goes to Backend
- Backend processes payment
- Payment details stored in Payments table
- Booking updated with payment status

Component Diagram (UML)



Components (Middle Part)

- Registration: handles new user sign-ups and account creation
- Authentication: manages login, identity verification, and user access
- Facility Management: manages sports facilities (courts, fields, availability, updates)
- Booking Management: handles reservations, schedules, and booking operations
- Staff Operations: manages staff roles, tasks, and internal operations
- Payment Interface: handles user payments and payment-related interactions

Left Side of the Diagram

- Sports Facility Booking System: represents the main system that connects all components together and serves as the central platform.

Data Access (Left Connections)

- All components are connected through a shared data flow
- Each module (Registration, Booking, Facility Management, etc.) can send and receive data
- This ensures consistent communication and proper data handling across the system

Right Side of the Diagram

- Security Management
- Ensures system protection through:
- Encryption: protects sensitive data (passwords, payment info)
- Access Control: ensures only authorized users can access certain features
- Additional services connected to security:
- Audit and Logging Service: records system activities for monitoring and tracking
- Backup and Recovery Service: ensures data safety and system recovery in case of failure

Persistence Service

- Handles storing and retrieving system data
- Acts as the bridge between application components and the database
- Also applies encryption and access control before storing data

Bottom Right Side

- Database: stores all system data (users, bookings, facilities, payments, staff info)
- Database Connector: manages communication between the persistence layer and the database
- Cache Service: temporarily stores frequently accessed data to improve performance and reduce database load

Bottom Middle (Payments)

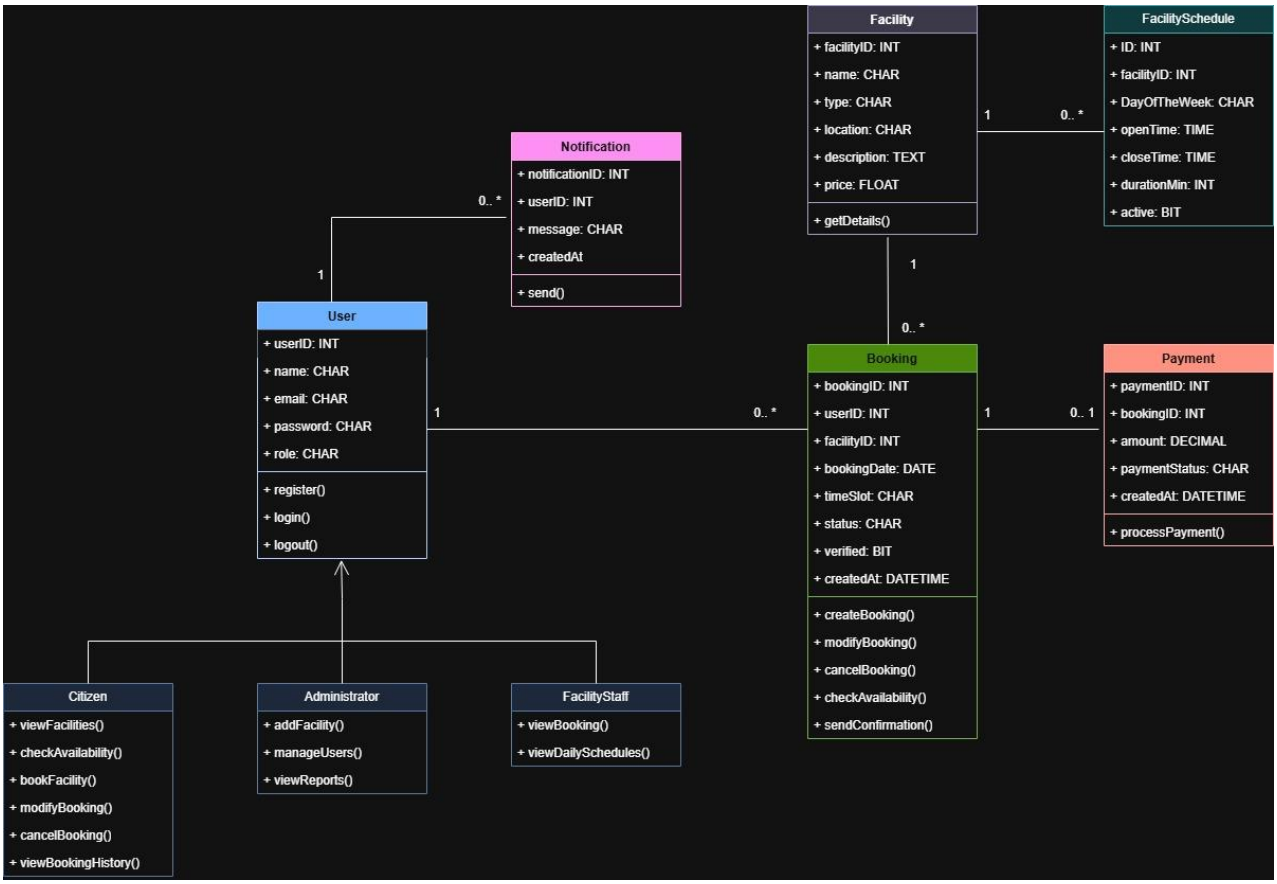
- Payment Gateway: processes payment transactions
- External Payment Provider: third-party service (bank or payment API) that completes the transaction

Interaction Explanation

- System - Components: the main system communicates with all modules (registration, booking, etc.)
- Components - Security Management: all sensitive data passes through security (encryption and access control)
- Components - Persistence Service: data is sent for storage or retrieval
- Persistence - Database: data is stored in the database
- Database Connector: manages communication between persistence and the database
- Cache Service: speeds up data access by storing frequently used data
- Payment Flow:
  Payment Interface → Payment Gateway → External Payment Provider

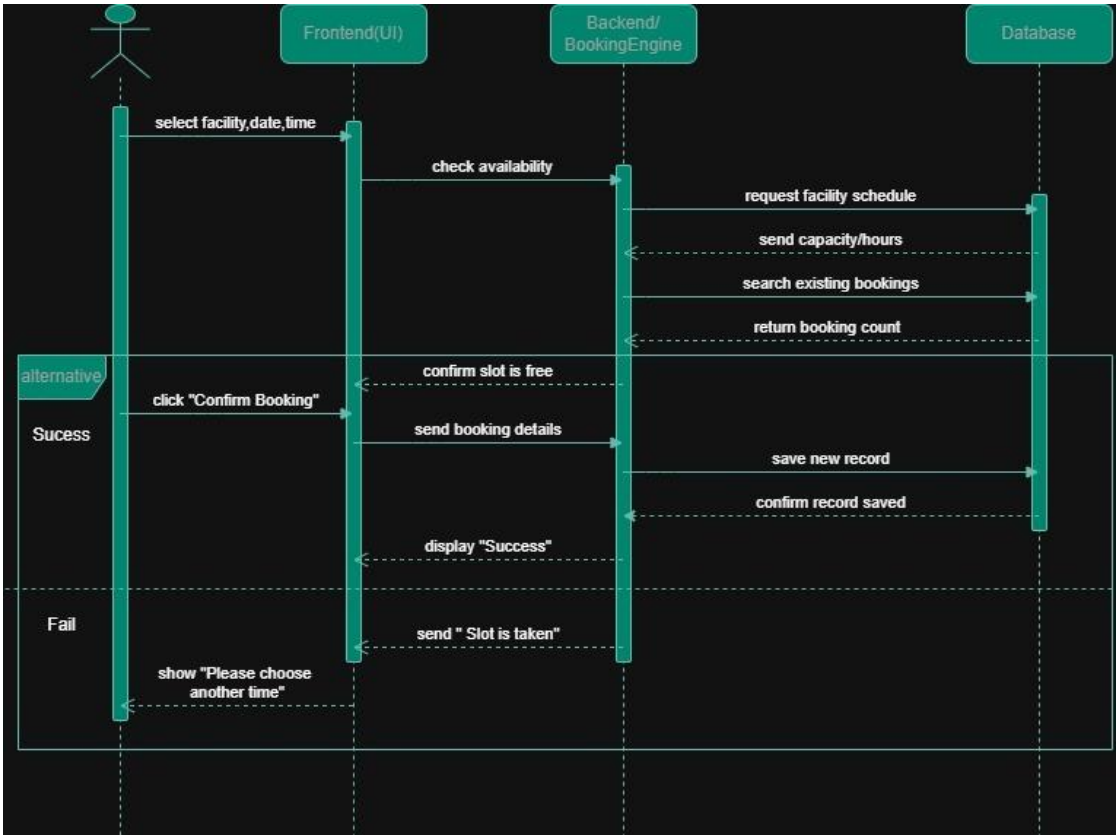
2. Detailed Design

Class Diagram

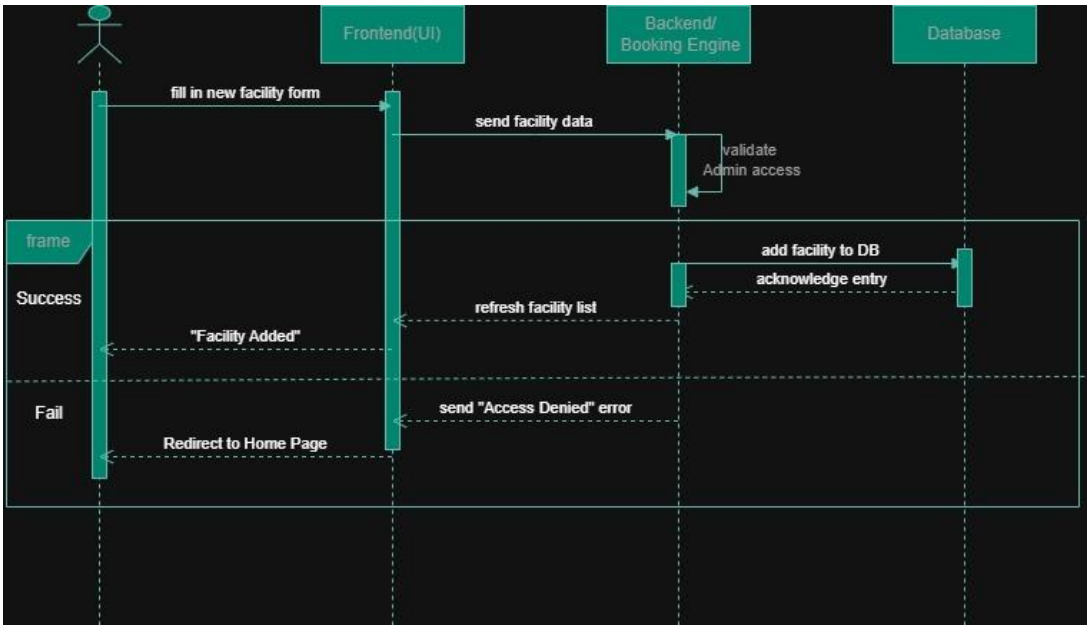


Sequence Diagrams

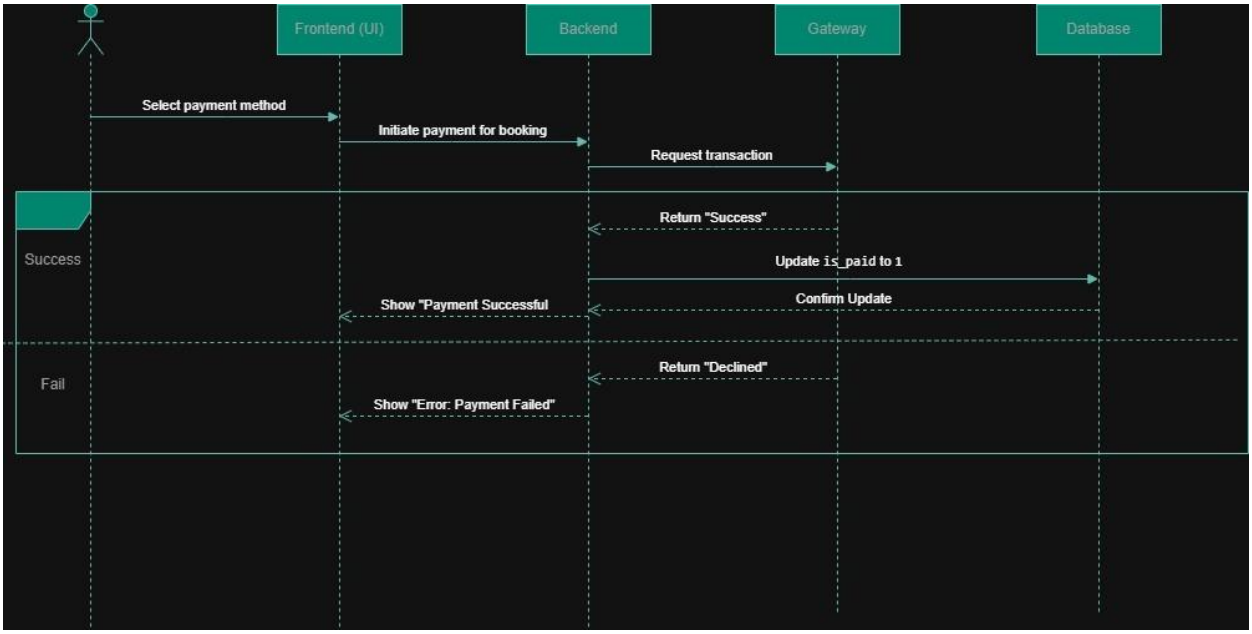
1. Facility Booking Sequence Diagram



2. Admin: Add Facility Sequence Diagram

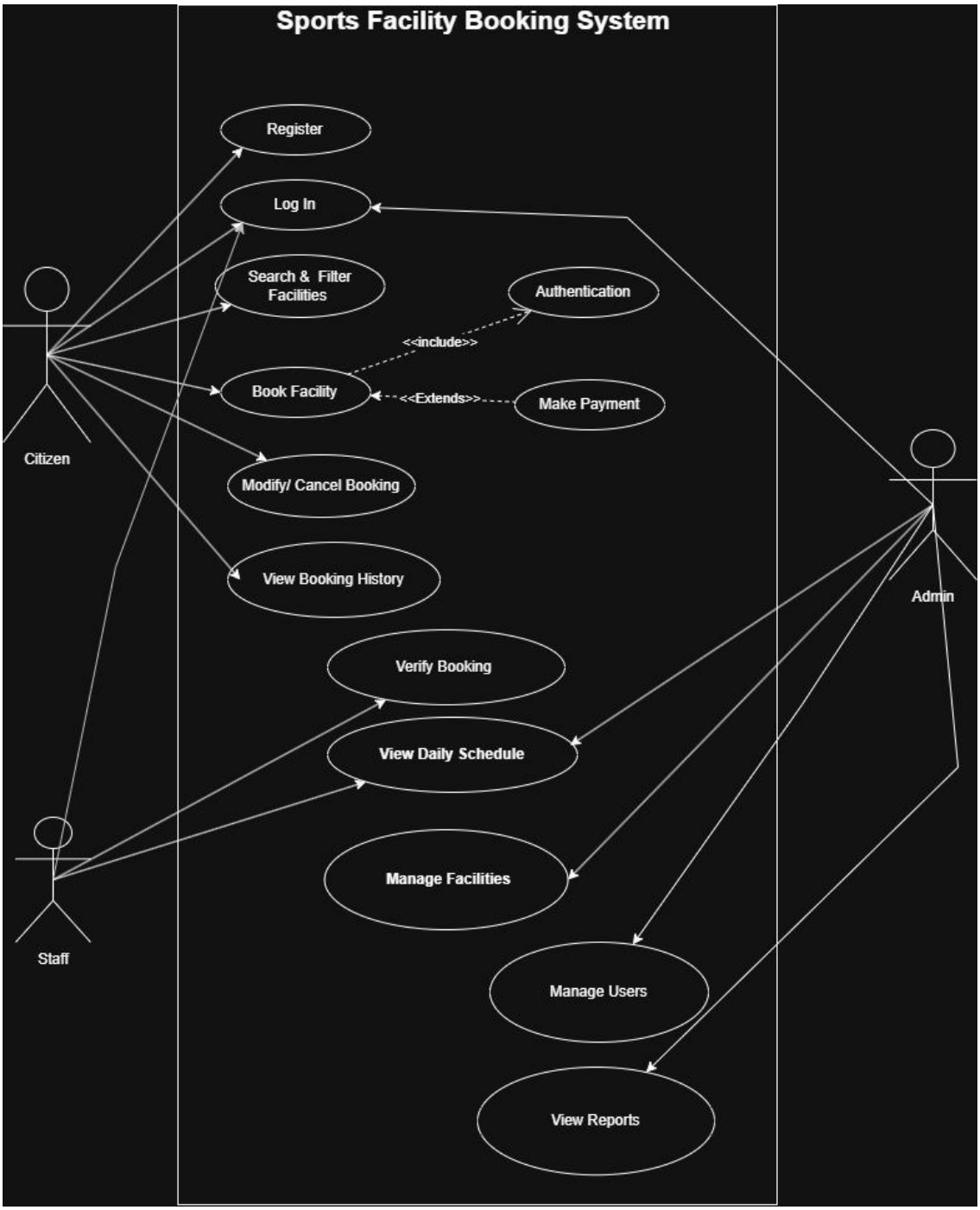


3. Optional Payment Sequence Diagram



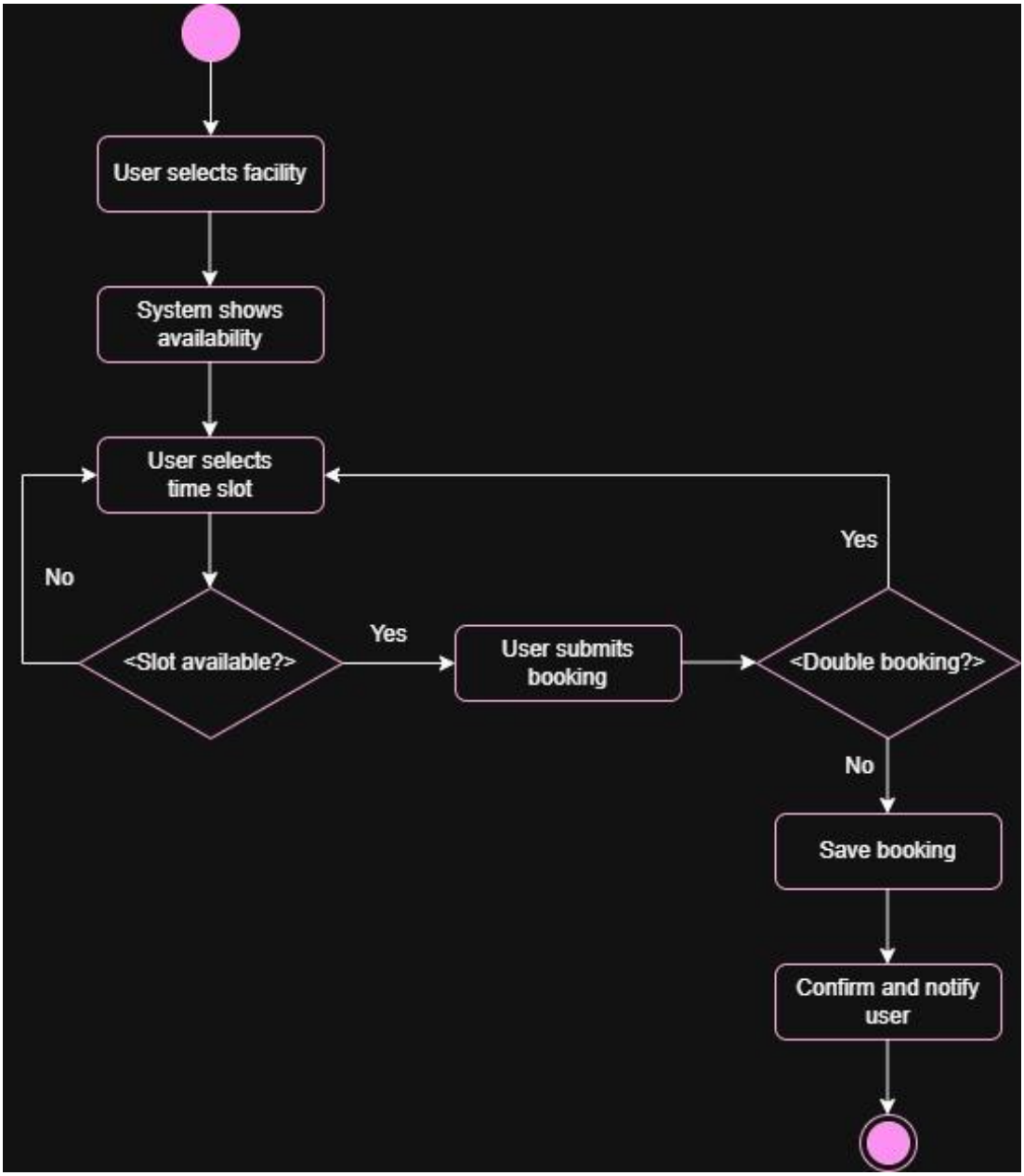
3. Modeling

Use Case Diagram



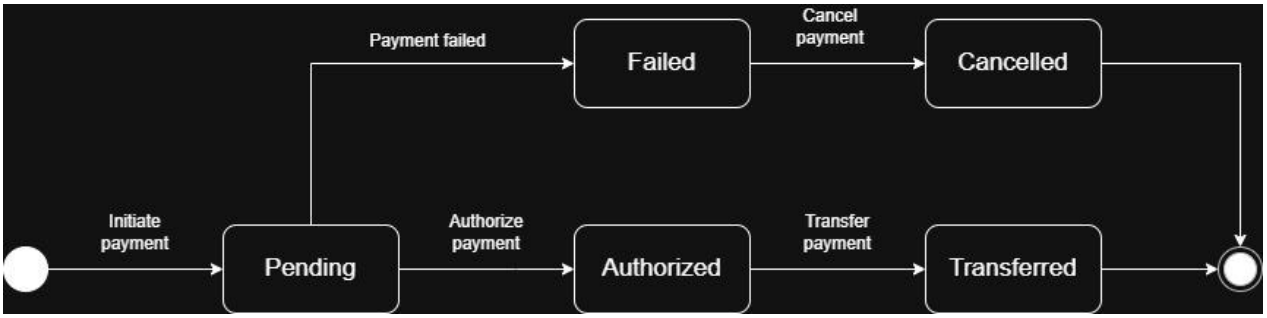
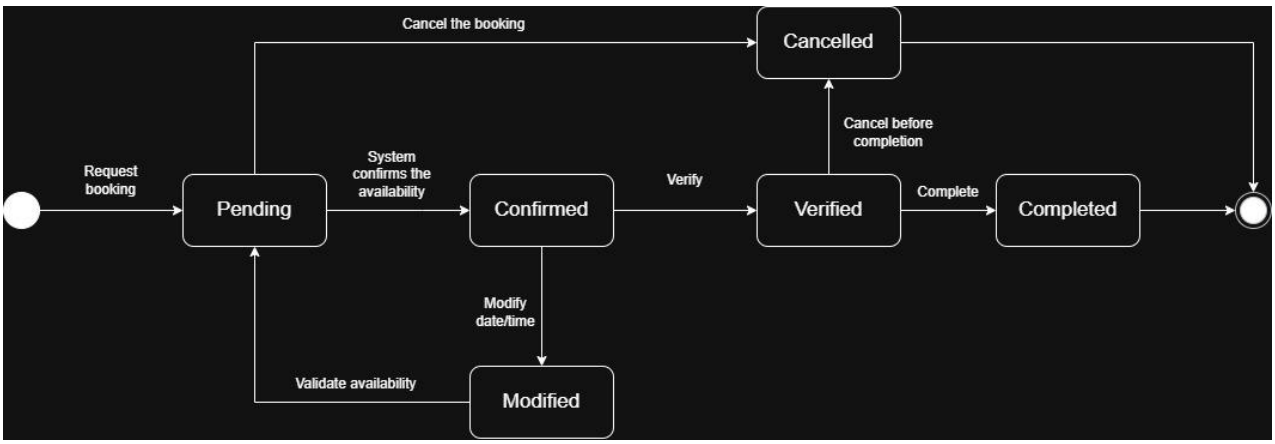
Activity Diagram

User Booking Process



State diagram

1. Payment Lifecycle



2. Booking Lifecycle